

Research Area Preferences (4 options)

Please choose up to four research areas. If you are interested in multiple areas of study, we strongly suggest that you apply to a higher number of research areas. That will help us better understand your interests and give you the greatest chance of being admitted to the program.

Research Areas offered:

Anthropology

Architecture

Art History

Astronomy

Biology

Business

Chemistry

Computer Science

Criminology

Ecology

Economics

Culture Studies

Education

Engineering

Environmental Studies

Film Studies

Gender Studies

History

International Relations

Literature

Mathematics

Media Studies

Music/Music Theory

Neuroscience

Physics

Political Science

Philosophy

Psychology

Science, Technology & Society (STS)*

Sociology

*STS is a multidisciplinary field that examines how social, cultural and political values influence and are influenced by scientific and technological developments.

Area of interest: Biology

1. Please list relevant courses and tests (SAT II, AP, etc.) that you've taken in this area. For STEM areas, please list and describe experience in calculus, statistics, other higher level math courses, chemistry, physics, biology, psychology, computer programming and coding (including scores and grades). (450 chars. maximum)

1. Science 98%
2. HL Biology (SBI3UH)
3. HL Chemistry (SCH3UH)
4. HL Math Analytics and Approaches (MHF4UH)

2. Do you have any other experience or qualifications in this field? Tell us anything about your background that may be helpful in demonstrating your interest in this specific subject area. Please answer in essay format. (450 chars. maximum)

I was introduced to biology in Grade 6 where I completed the Forensic Science course offered by CTY at JHU. I competed in USABO (silver), HOSA (top 15%), and iGEM (silver, team leader). I volunteered to connect with patients and explore multiple aspects of biology at Beijing Harmony Hospice, STIA Elderly Care, and Shanghai United Family Hospital. I also joined summer programs including JHU Medical School Intensive and ASDAN Neuroscience.

3. Why do you want to engage in this research area? Please answer in essay format. (100-150 words)

Having a solid biology foundation is essential for a future medical career. When volunteering at the Hospice program in Beijing Harmony Hospital, I know that treating illnesses at the late stage of the patients' lives is stressful for many families. I have seen elders who cannot move lying on the bed to receive treatments. It is undoubtedly excruciating for them. Therefore, a solid knowledge of the patient's medical concerns and proposed solutions reduce the angst for the patient and their loved ones. Biology research will prepare me to learn more about diseases and possible solutions. I genuinely hope that in the future, through the contribution of myself and countless others, that medical solutions are accessible to all.

Area of interest: Neuroscience

1. Please list relevant courses and tests (SAT II, AP, etc.) that you've taken in this area. For STEM areas, please list and describe experience in calculus, statistics, other higher level math courses, chemistry, physics, biology,

psychology, computer programming and coding (including scores and grades).
(450 chars. maximum)

1. HL Biology (SBI3UH)
2. HL Chemistry (SCH3UH)
3. HL Math Analytics and Approaches (MHF4UH)

2. Do you have any other experience or qualifications in this field? Tell us anything about your background that may be helpful in demonstrating your interest in this specific subject area. Please answer in essay format. (450 chars. maximum)

In the summer before rising into Grade 7, I attended an introductory Neuroscience summer school. In summer 2022, I attended an advanced Neuroscience course provided by ASDAN, learning about frequently used databases such as Allen Brain and Kegg while also studying neuroscience research methods. I have also competed in USABO, iGEM, and HOSA, volunteered at hospitals, and participated in summer schools including JHU Medical School Intensive.

3. Why do you want to engage in this research area? Please answer in essay format. (100-150 words)

My volunteer service includes care-taking of the elderly who are suffering from Alzheimer's Disease. I have witnessed the harrowing experience of losing one's memory which has inspired me to seek solutions. I want to commit my career to actualize cures for common neurological disorders such as Alzheimer's through research.

While it is not currently possible to cure Alzheimer's, I intend to research to cure this fatal disease. Neuroscience is fundamental not only to Alzheimer's but also navigation of other neurological disorders that strike humans. It is estimated that there are globally 47.5 million people with dementia, with 7.7 million new cases every year. Alzheimer's disease is the most common cause of dementia and may contribute to 60–70% of cases. My eventual research in neuroscience will hopefully help me heal those stricken with neurological disorders so that these patients can lead fulfilling lives.

Application essay (required):

"Obstacles do not bend me. Every obstacle is destroyed through rigor." -
Leonardo da Vinci

The Pioneer Research Program is one of the most rigorous and challenging

academic endeavors a student can undertake during high school. How will Pioneer's rigorous system empower you to exceed your expectations of yourself? (We **encourage** you to reference ideas on our website or specific information about Pioneer that you may have learned from others if applicable.) (Please compose a 300-400 word essay)*

Challenges and growth are two themes of my academic life. Given how selective the Pioneer Research Program is, I hope to achieve profound academic growth by overcoming rigorous challenges.

The abundant academic resources of the program are the most appealing. As many alumni mentioned, a highlight of the Pioneer Research program is individual research mentoring. One of the challenges I had when participating in iGEM was needing more specific feedback from our mentors as we were severely short-staffed.

However, in Pioneer, when connected with specialized mentors, I will delve into the topics I am passionate about, such as learning more about the biology behind certain diseases and possible solutions to cure Alzheimer's. Academic support provided by the program to help first-time researchers like me is inspiring. Shannon Heh, an alumnus of the program, has mentioned that there are workshops and writing centers to guide students through the process of producing an undergraduate-level capstone project. These resources will not only help me write a research paper but also develop a foundation for any of my future research opportunities.

Besides my desire to grow, I also possess the ability to overcome challenges. At school, besides committing to the most rigorous courses, I also sought outside opportunities to join academic challenges like iGEM. When I first joined, I did not stand out in the team. However, over the course of the competition, my instructor and peers witnessed the effort I exerted and the initiative I exemplified to lead the team. Eventually, they elected me as the team leader to guide our group throughout the rest of the competition. I challenged myself to go beyond my limits, as overcoming challenges is the source of growth. At Pioneer, I will have the opportunity to challenge myself once again and explore my capacity.

By applying to the Pioneer Research Program, I aim to exceed my limits with the support of excellent academic resources. Specifically, I hope to gain more knowledge on biology and neuroscience to help facilitate my future exploration of combating Alzheimer's so that the elderly can lead respectful and dignified lives.